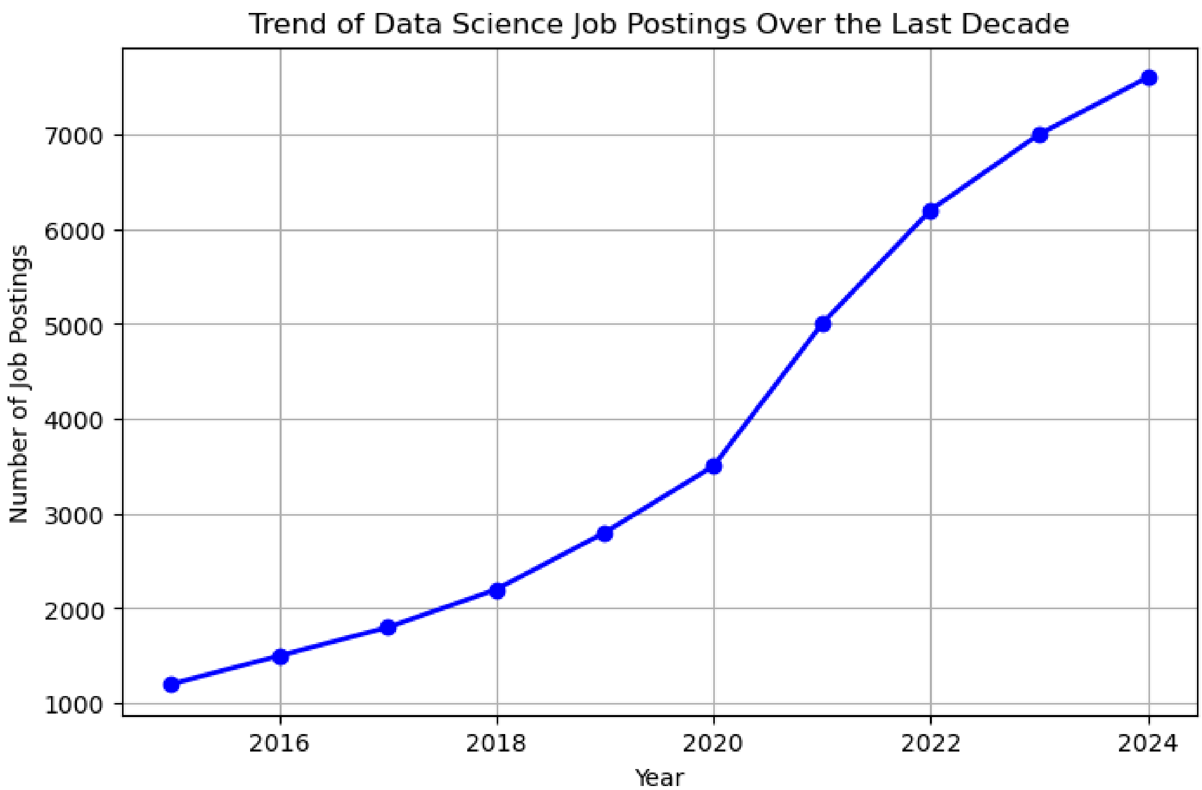
In [ ]:

```
1.a. Analyze the trend of data
science job postings over the last decade for the given  data on the number of data
postings each year. Use pandas for data manipulation and matplotlib  for visualizat
```

In [2]:

```python
import pandas as pd
import matplotlib.pyplot as plt

# Sample data (replace with your given data)
data = {
    'Year': [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024],
    'Job_Postings': [1200, 1500, 1800, 2200, 2800, 3500, 5000, 6200, 7000, 7600]
}

# Create DataFrame
df = pd.DataFrame(data)

# Display the data
print(df)

# Plot the trend
plt.figure(figsize=(8,5))
plt.plot(df['Year'], df['Job_Postings'], marker='o', color='blue', linewidth=2)

# Add labels and title
plt.title('Trend of Data Science Job Postings Over the Last Decade')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.grid(True)

# Show the plot
plt.show()
```

```
   Year  Job_Postings
0  2015          1200
1  2016          1500
2  2017          1800
3  2018          2200
4  2019          2800
5  2020          3500
6  2021          5000
7  2022          6200
8  2023          7000
9  2024          7600
```

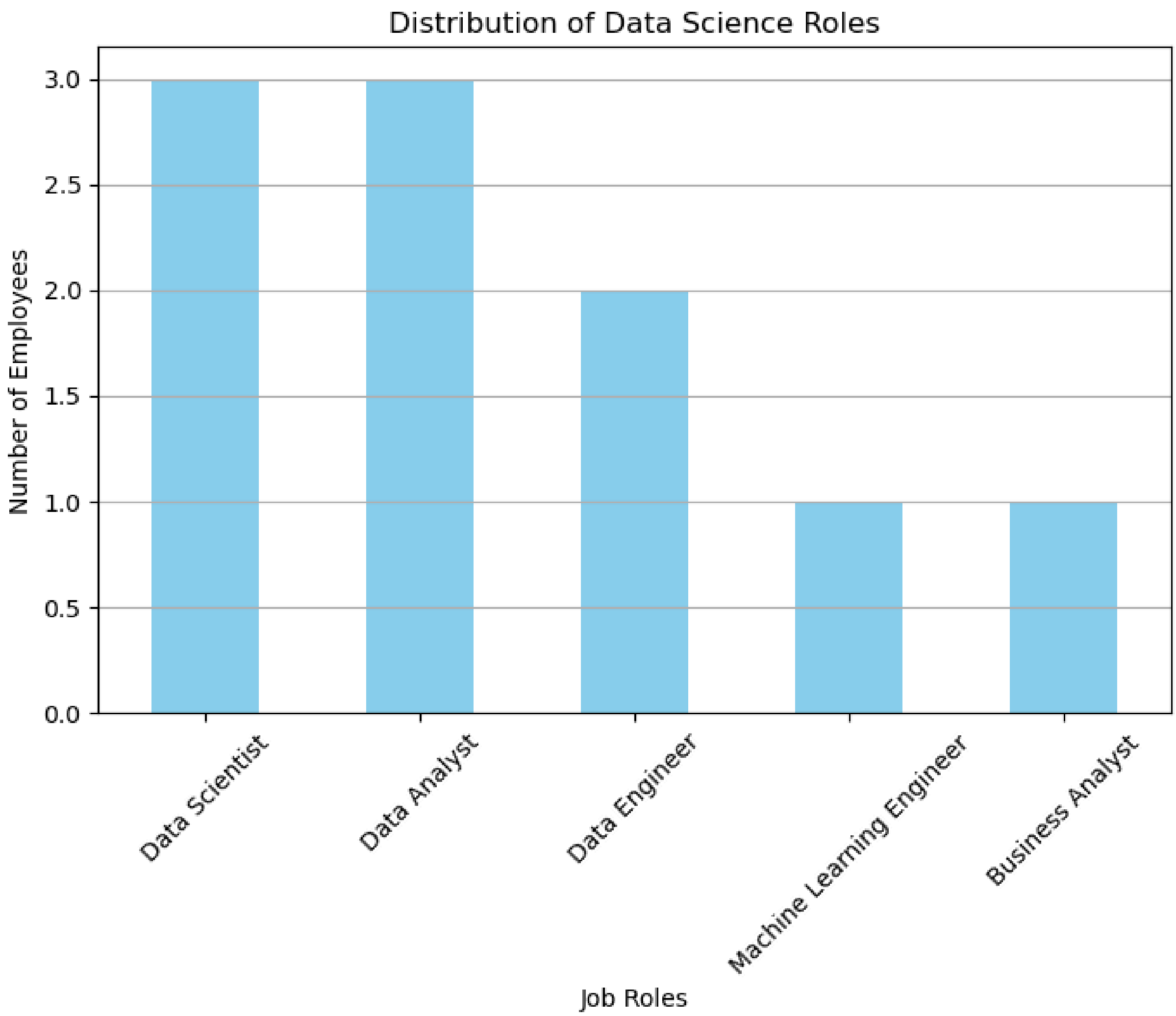


1.b.Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

In [3]:
```
import pandas as pd
import matplotlib.pyplot as plt

# Sample dataset
data = {
    'Role': [
        'Data Scientist', 'Data Analyst', 'Data Engineer',
        'Data Scientist', 'Data Analyst', 'Machine Learning Engineer',
        'Data Engineer', 'Data Scientist', 'Business Analyst',
        'Data Analyst'
    ]
}

# Create DataFrame
df = pd.DataFrame(data)

# Count the frequency of each role
role_counts = df['Role'].value_counts()

# Display the counts
print(role_counts)

# Plot the bar chart
plt.figure(figsize=(8,5))
role_counts.plot(kind='bar', color='skyblue')

plt.title('Distribution of Data Science Roles')
```

```
plt.xlabel('Job Roles')
plt.ylabel('Number of Employees')
plt.xticks(rotation=45)
plt.grid(axis='y')

plt.show()
```

```
Role
Data Scientist              3
Data Analyst                3
Data Engineer               2
Machine Learning Engineer   1
Business Analyst            1
Name: count, dtype: int64
```



1.c.Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

In [4]:
```
import pandas as pd
import json

# -------------------------------
# Structured Data
# -------------------------------
```

```python
structured_data = {
    "ID": [101, 102, 103],
    "Name": ["Alice", "Bob", "Charlie"],
    "Age": [24, 28, 22]
}

structured_df = pd.DataFrame(structured_data)

print("Structured Data")
print(structured_df)

# ------------------------------
# Semi-Structured Data (JSON)
# ------------------------------
semi_structured_data = [
    {"ID": 101, "Name": "Alice", "Skills": ["Python", "SQL"]},
    {"ID": 102, "Name": "Bob", "Skills": ["Java", "C++"]},
    {"ID": 103, "Name": "Charlie", "Skills": ["Power BI", "Excel"]}
]

print("\nSemi-Structured Data")
print(json.dumps(semi_structured_data, indent=4))

# ------------------------------
# Unstructured Data
# ------------------------------
unstructured_data = """
Data Science is an interdisciplinary field that uses scientific methods,
machine learning, and statistics to extract knowledge from data.
"""

print("\nUnstructured Data")
print(unstructured_data)
```

```
Structured Data
    ID     Name  Age
0  101    Alice   24
1  102      Bob   28
2  103  Charlie   22

Semi-Structured Data
[
    {
        "ID": 101,
        "Name": "Alice",
        "Skills": [
            "Python",
            "SQL"
        ]
    },
    {
        "ID": 102,
        "Name": "Bob",
        "Skills": [
            "Java",
            "C++"
        ]
    },
    {
        "ID": 103,
        "Name": "Charlie",
        "Skills": [
            "Power BI",
            "Excel"
        ]
    }
]

Unstructured Data

Data Science is an interdisciplinary field that uses scientific methods,
machine learning, and statistics to extract knowledge from data.
```

1.d.Conduct an experiment to encrypt and decrypt given sensitive data using the cryptography library fernet.

In [5]:

```python
from cryptography.fernet import Fernet

# Step 1: Generate a secret key
key = Fernet.generate_key()

# Step 2: Create a Fernet object using the key
cipher = Fernet(key)

# Step 3: Original sensitive data
message = "My ATM PIN is 1234"

print("Original Message:")
```

```
print(message)

# Step 4: Encrypt the message
encrypted_message = cipher.encrypt(message.encode())

print("\nEncrypted Message:")
print(encrypted_message)

# Step 5: Decrypt the message
decrypted_message = cipher.decrypt(encrypted_message)

print("\nDecrypted Message:")
print(decrypted_message.decode())
```

```
Original Message:
My ATM PIN is 1234

Encrypted Message:
b'gAAAAABqcJYW_OEuR5wkvHRhxI1KUV-OxQ_DTAZSNJyS0rHXWzWjYZHZM3Cm5n7Jq3hUr7NGgOeFqDjLfJ
QMFNbIsLc8MXJOjHilMshXZuq72Uz4NHYzXoY='

Decrypted Message:
My ATM PIN is 1234
```

In [ ]:

```

```